

Electricity and Magnetism

1. Introduction

Electricity and magnetism form one of the most important branches of physics because they explain electric charge, electric fields, magnetic fields, circuits, and electromagnetic phenomena. In BSc Physics, this subject develops the mathematical and physical foundation for devices, power systems, communication technology, and modern electronics.

The subject is usually studied in a sequence that begins with vector analysis and electrostatics, then moves through dielectrics, current electricity, magnetic effects of current, electromagnetic induction, and alternating current.

2. Electric charge

Electric charge is a fundamental property of matter responsible for electric interactions. It exists in two kinds, positive and negative, and is **quantized**, meaning it occurs in integral multiples of the elementary charge e .

Properties of charge

- Charge is conserved.
- Charge is additive.
- Charge is quantized.
- Like charges repel and unlike charges attract.

3. Coulomb's law

Coulomb's law gives the electrostatic force between two-point charges. It is one of the core formulas of electricity and magnetism.

Formula

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

where q_1 and q_2 are charges and r is the distance between them.

Important points

- The force acts along the line joining the charges.
- It obeys the inverse-square law.
- In a medium, ϵ_0 is replaced by ϵ .

4. Electric field

The electric field is the region in which a charge experiences force. It is a vector quantity.

Field definition

$$\mathbf{E} = \frac{\mathbf{F}}{q}$$

where q is the test charge.

Field due to a point charge

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Field lines

- Start on positive charges and end on negative charges.
- Never intersect.
- Density of lines indicates field strength.

5. Electric flux and Gauss's law

Electric flux measures the number of field lines passing through a surface.

Flux

$$\Phi_E = \int \mathbf{E} \cdot d\mathbf{A}$$

Gauss's law

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

This law is especially useful for symmetric charge distributions.

Applications

- Infinite line charge.
- Infinite plane sheet.
- Spherical charge distribution.

6. Electric potential

Electric potential is the work done per unit charge in bringing a test charge from infinity to a point in the field.

Potential

$$V = \frac{W}{q}$$

Potential due to a point charge

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Relation between field and potential

$$\mathbf{E} = -\nabla V$$

or in one dimension,

$$E = -\frac{dV}{dx}$$

7. Electrostatic potential energy

The electrostatic potential energy is the work required to assemble a system of charges.

For two-point charges

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

8. Capacitance

A capacitor stores charge and electrical energy. Capacitance is the ability to store charge per unit potential difference.

Definition

$$C = \frac{Q}{V}$$

Parallel plate capacitor

$$C = \epsilon_0 \frac{A}{d}$$

With dielectric,

$$C = \kappa\epsilon_0 \frac{A}{d}$$

Energy stored

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$$

9. Dielectrics and polarization

Dielectrics are insulating materials that become polarized in an electric field. Polarization reduces the effective field and increases capacitance in a capacitor.

Important terms

- Polarization.
- Dielectric constant κ .
- Electric susceptibility.

10. Current electricity

Current electricity deals with the flow of charge through conductors. It includes current, resistance, resistivity, Ohm's law, circuits, and electrical power.

Current

$$I = \frac{dq}{dt}$$

Ohm's law

$$V = IR$$

where R is resistance.

Resistance

$$R = \rho \frac{L}{A}$$

where ρ is resistivity.

Conductance

$$G = \frac{1}{R}$$

Power

$$P = VI = I^2R = \frac{V^2}{R}$$

11. Kirchhoff's laws

Kirchhoff's laws are used to analyze complex circuits.

Junction rule

The algebraic sum of currents at a junction is zero.

Loop rule

The algebraic sum of potential differences in a closed loop is zero.

12. Wheatstone bridge and meter bridge

These bridge circuits are used to measure unknown resistance precisely.

Wheatstone bridge balance condition

$$\frac{P}{Q} = \frac{R}{S}$$

13. Drift velocity and mobility

In conductors, electrons have random motion, but an electric field produces a net drift.

Current in terms of drift speed

$$I = nAev_d$$

where n is carrier density and v_d is drift velocity.

Mobility

$$\mu = \frac{v_d}{E}$$

14. Magnetic field

Magnetic fields are produced by moving charges and currents. A moving charge in a magnetic field experiences a force perpendicular to both velocity and field.

Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Force on a moving charge in magnetic field

$$F = qvB\sin\theta$$

15. Biot-Savart law

The Biot-Savart law gives the magnetic field due to a current element.

Formula

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \hat{r}}{r^2}$$

Applications

- Field due to long straight wire.
- Circular coil.
- Solenoid.

16. Ampere's law

Ampere's circuital law is useful for symmetric current distributions.

Formula

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

17. Magnetic materials

Materials respond differently to magnetic fields and are classified as diamagnetic, paramagnetic, or ferromagnetic.

Important terms

- Magnetization.
- Magnetic susceptibility.
- Permeability.
- Hysteresis.

18. Electromagnetic induction

Changing magnetic flux induces an emf. This is the basis of generators, transformers, and many electrical machines.

Faraday's law

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Lenz's law

The induced emf opposes the change in magnetic flux that produces it.

19. Self and mutual inductance

Inductance arises because changing current produces changing magnetic flux.

Self-inductance

$$\mathcal{E} = -L \frac{dI}{dt}$$

Energy in inductor

$$U = \frac{1}{2} LI^2$$

20. Alternating current

Alternating current changes periodically with time. AC theory includes reactance, impedance, resonance, and power factor.

RMS values

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}, V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

Capacitive reactance

$$X_C = \frac{1}{\omega C}$$

Inductive reactance

$$X_L = \omega L$$

Impedance of RLC circuit

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Power in AC

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

21. Resonance in AC

Resonance occurs when $X_L = X_C$. At resonance, impedance is minimum and current is maximum in a series RLC circuit.

Resonant frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

22. Maxwell's equations

Maxwell's equations unify electricity and magnetism into electromagnetism. They are the fundamental equations of the field.

In integral form

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\oint \mathbf{B} \cdot d\mathbf{A} = 0$$

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

23. Important formula sheet

- $\mathbf{E} = \mathbf{F}/q$.
- $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$.
- $\Phi_E = \int \mathbf{E} \cdot d\mathbf{A}$.
- $\oint \mathbf{E} \cdot d\mathbf{A} = q_{\text{enc}}/\epsilon_0$.
- $V = W/q$.
- $C = Q/V$.
- $V = IR$.
- $P = VI = I^2 R = V^2/R$.
- $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- $\mathcal{E} = -d\Phi_B/dt$.
- $\mathcal{E} = -L di/dt$.
- $I_{\text{rms}} = I_0/\sqrt{2}$.
- $Z = \sqrt{R^2 + (X_L - X_C)^2}$.
- $\omega_0 = 1/\sqrt{LC}$.

24. Exam-oriented distinctions

Conductor vs dielectric

- Conductors allow free charge flow.
- Dielectrics oppose free charge motion but polarize in an electric field.

Electric field vs electric potential

- Electric field is force per unit charge.
- Potential is work per unit charge.

Magnetic field vs electric field

- Electric fields act on charges even at rest.
- Magnetic fields act only on moving charges.

Self-induction vs mutual induction

- Self-induction is induced emf in the same coil.
- Mutual induction is induced emf in a neighboring coil.

25. Summary

Electricity and magnetism is a foundational BSc Physics subject that connects electrostatics, current electricity, magnetism, induction, and alternating currents into a unified framework. It is essential for understanding circuits, measurement, motors, generators, transformers, and the deeper Maxwellian picture of electromagnetism.